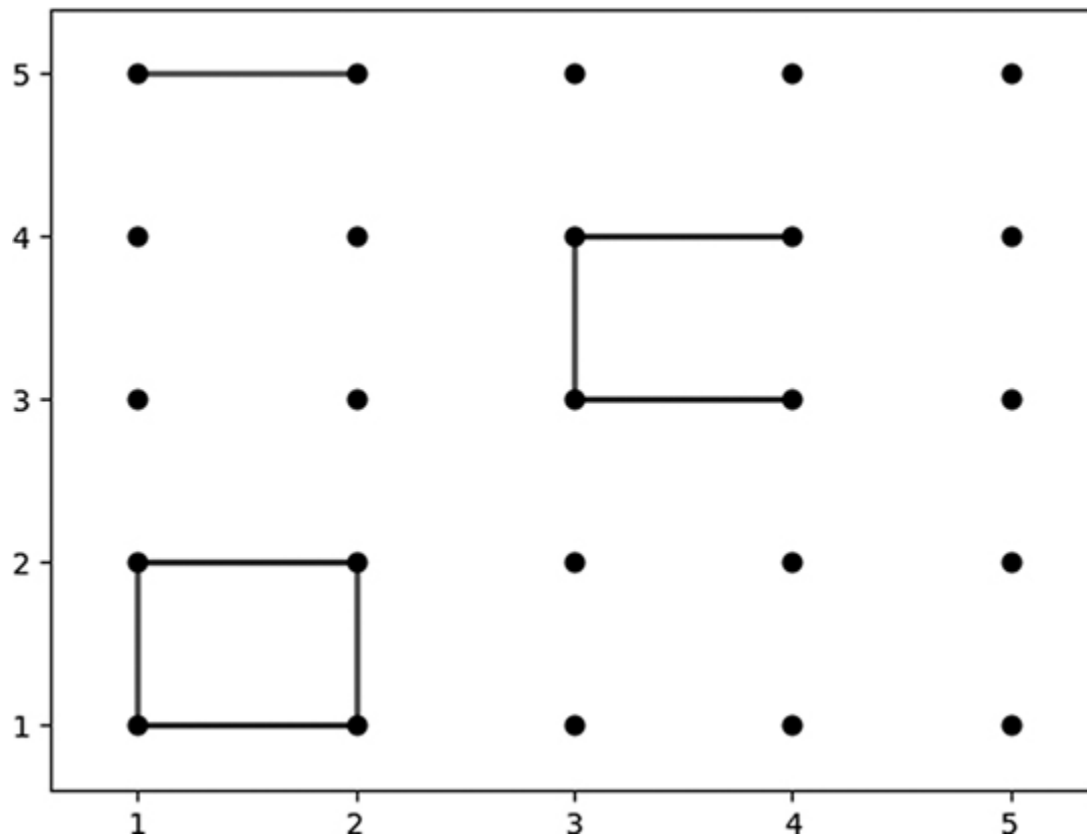


and earn a point. Players only earn points for completing the smallest possible squares on the board: those with a side length of 1. The player who's earned the most points when the lattice is completely filled in with line segments wins the game. There are some variations on the game, including different board shapes and more advanced rules, but the simple AI we'll build in this chapter will work with the rules we've described here.



[Figure 10-2](#): A dots and boxes game in progress

Drawing the Board

Though not strictly necessary for our algorithmic purposes, drawing the board can make it easier to visualize the ideas we're discussing. A very simple plotting function can make an $n \times n$ lattice by looping over x and y coordinates and using the `plot()` function in Python's `matplotlib` module:

```
import matplotlib.pyplot as plt
from matplotlib import collections as mc
def drawlattice(n,name):
```